

Elham Esmaeilnia

Senior AI Research & Development Engineer

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Tehran, Iran

Senior AI Engineer and Researcher with experience in Artificial Intelligence, Machine Learning, NLP, and Big Data systems. Specialized in Large Language Models (LLMs), Speech processing, Intelligent agents, and Decision-support systems. Background in both academic research and industrial AI development.

Work Experience

Senior AI Engineer – Parnian Gostar

Aug 2025 – Jan 2026

- Design and development of AI-based Financial model AlphaFusionNet
- Implementation of LLM-based financial intelligent agents
- Development of timeseries pipelines and deep learning models
- Building AI-driven hybrid decision-making systems

AI Research & Development Engineer – Bertina

Dec 2024 – 2025

- Research and development of Persian NLP systems
- Contributor and co-author of the G2P article from the Bertix Artificial Intelligence Group.
- Designing text normalization, tokenization for G2P models
- Working on language understanding systems
- Designing a diacritization model for converting raw Persian texts into Persian texts with diacritics

AI Research & Development Engineer – Farajahan

Nov 2023 – Dec 2024

- Development of intelligent QA and text correction systems
- Developed a system to generate intelligent responses from Persian content using RAG.
- Development of Persian speech enhancement model
- Training transformer-based language models and Research on Persian NLP applications
- Created a Persian intelligent assistant leveraging LLM and Crewai to perform multiple task-based functions.

Big Data & Web Developer – Kanda Idea

Jun 2020 – Aug 2022

- Big data processing using Spark, Hadoop, and HDFS
- Backend and API development
- Data analytics and pipeline implementation using Cassandra
- Web-based software programming (IMIDRO, WEB MARINE)
- Design use-case and sequence diagrams for software architecture

AI Researcher – IPM

Apr 2018 – Aug 2018

- Research on Deep learning and semi-supervised processing

Instructor – Alavi Institute

Jun 2013 – Jul 2014

- Teaching programming, computer fundamentals and Mathematics

Education

PhD Candidate in Computer Engineering (AI & Robotics)

Islamic Azad University – Tehran Central

MSc in Computer Engineering (Artificial Intelligence)

Rouzbehan University_ Sari

BSc in Computer Engineering

Shahid Beheshti University_ Tehran

Technical Skills

Programming: Python, C++, MATLAB, JavaScript/JQuery, HTML/CSS, VHDL

AI & ML: TensorFlow, PyTorch, Keras, Transformers, Stable-Baselines3, kerasRL, Librosa, Audiolib, scikit-learn, Opencv, ...

NLP & LLMs & Agentics & Timeseries: LangChain, Langgraph, CrewAI, RAG, vectorDB, Finrobot, Bigbird, CGA, BERT, T5, LSTM, CNN,

Big Data: Spark, Hadoop, HDFS, Cassandra, Neo4j, MongoDB, Redis, ..

Tools: Docker, Git, Celery, IBM spss Modeler, RapidMiner

Selected Projects

- AI-based Financial Portfolio Optimization System
 - Designing a Diacritization model for converting raw Persian texts into Persian texts with diacritics
 - Persian NLP Pipeline (G2P, Tokenization, Text Correction)
 - RAG-based Intelligent Assistant using LLMs
 - Speech Enhancement using Deep Learning
 - Created a Persian intelligent assistant leveraging LLM and Crewai to perform multiple task-based functions.
 - Web-based systems, IMIDRO & WEB MARINE projects
 - Human behavior recognition in video using sparse coding
 - Video Forgery detection & Facial Expression Recognition
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Certifications

- Data Science & Python – DigiNext
- Web Design – FanAsa
- Teaching Skills – Ministry of Education
- IT Internship – Agricultural Jihad Organization

Honors & Achievements

- National Mathematics Olympiad – Selected Participant
- City-Level Mathematics Olympiad – Selected Participant
- City-Level Physics Olympiad – Selected Participant

Languages

- Persian (Native)
- English (Professional)